

A lower bound of 1.373 for the complex Grothendieck constant

Abstract

We prove that the classical complex Grothendieck constant exceeds 1.373. The construction subtracts an identity term and the full real third-chaos projection from a holomorphic Gaussian first-chaos projection. An anisotropic multiplication inequality bounds the cubic projection in every dimension. An exact pointwise formula uses the orthogonality of the two midpoint functions associated with complex phases. The remaining scalar parameter is controlled on its entire range by convex tangent bounds. A finite family of auxiliary rational step weights supplies different bounds on prescribed parameter intervals for one fixed operator. Exact polynomial exponential moments and an explicit finite matrix complete the proof.

1 The constant and the result

For a nonzero complex m -by- n matrix $A = (a_{ij})$, let

$$S_{\mathbb{C}}(A) = \max_{|\alpha_i|=|\beta_j|=1} \left| \sum_{i,j} a_{ij} \alpha_i \overline{\beta_j} \right|,$$
$$V_{\mathbb{C}}(A) = \sup_{d \geq 1} \max_{\|u_i\|=\|v_j\|=1} \left| \sum_{i,j} a_{ij} \langle u_i, v_j \rangle \right|,$$

where all vectors belong to $\mathbb{C}^d$ and $\langle u, v \rangle = \sum_r u_r \overline{v_r}$. The classical complex Grothendieck constant is $K_G^{\mathbb{C}} = \sup_{m,n,A \neq 0} V_{\mathbb{C}}(A)/S_{\mathbb{C}}(A)$.

Theorem 1. *There is an explicitly specified finite complex matrix M such that*

$$\frac{V_{\mathbb{C}}(M)}{S_{\mathbb{C}}(M)} > 1.373.$$

Consequently, $K_G^{\mathbb{C}} > 1.373$.

The classical lower bound approximately 1.33807 is attributed to Davie in Haagerup [1] and Friedland–Lim–Zhang [2]. Haagerup proves an upper bound approximately 1.40491. Theorem 1 reduces the width of this reported interval by approximately 52 percent. Heilman [3, Section 1.8] proposes analysis of a cubic perturbation as a possible improvement in the complex case and leaves that analysis open. Here the operator is specified with its full real third-chaos projection, and its scalar bound covers every measurable complex phase assignment. The proof is independent of any unpublished lower-bound argument.

2 The Gaussian operator and its parameters

Let γ_d have density $\pi^{-d}e^{-\|z\|^2}$ on $\mathbb{C}^d$, and use a first-linear inner product on $L^2(\gamma_d; \mathbb{C})$. Let P be orthogonal projection onto $\text{span}_{\mathbb{C}}\{z_1, \dots, z_d\}$. The real coordinates

$$w = (\sqrt{2} \operatorname{Re} z_1, \sqrt{2} \operatorname{Im} z_1, \dots, \sqrt{2} \operatorname{Re} z_d, \sqrt{2} \operatorname{Im} z_d)$$

are independent standard Gaussians. For a standard real Gaussian coordinate, define the normalized Hermite polynomials by

$$H_r(x) = \frac{(-1)^r}{\sqrt{r!}} e^{x^2/2} \frac{d^r}{dx^r} e^{-x^2/2}.$$

The complex span of the products $\prod_{\ell=1}^{2d} H_{a_\ell}(w_\ell)$ with $\sum_{\ell} a_\ell = j$ is denoted by $\mathcal{H}_j(\mathbb{C}^d)$. Let C_3 be orthogonal projection onto $\mathcal{H}_3(\mathbb{C}^d)$. In particular, C_3 is the complexification of the full real third-chaos projection. The projections P, C_3 commute with unitary changes of the complex coordinates.

Throughout the construction the parameters are

$$\lambda = \frac{53326}{10^6}, \quad \beta = \frac{499883}{10^6}, \quad D = \frac{3447}{5000}. \quad (1)$$

Set

$$T = P - \lambda I - \beta C_3. \quad (2)$$

Its scalar norm is

$$\|T\|_{\infty \rightarrow 1} = \sup_{\|f\|_{\infty}, \|g\|_{\infty} \leq 1} |\langle f, Tg \rangle|.$$

Below we prove the dimension-independent bound $\|T\|_{\infty \rightarrow 1} \leq D$. Later sections construct a finite matrix with scalar value at most D and vector value greater than $1.373D$.

3 An anisotropic multiplication inequality

Write $z_1 = \sqrt{s}e^{i\phi}$, choosing ϕ arbitrarily on the null set $z_1 = 0$. For real functions of s , use the notation $E_s[F] = \int_0^\infty F(s)e^{-s} ds$. Let $q_R(s) \geq q_I(s) > 0$ be measurable weights, and put

$$\mathcal{Q}(k) = \int \left[q_R(s)(\operatorname{Re}(e^{-i\phi}k))^2 + q_I(s)(\operatorname{Im}(e^{-i\phi}k))^2 \right] d\gamma_d, \quad (3)$$

$$a(s) = \frac{1}{2} (q_R(s)^{-1} + q_I(s)^{-1}), \quad e(s) = \frac{1}{2} (q_R(s)^{-1} - q_I(s)^{-1}). \quad (4)$$

Define

$$p_4(s) = s^3/6, \quad p_5(s) = s(s-2)^2/2 \quad (5)$$

and the symmetric matrices

$$G_2 = \begin{pmatrix} E_s[a(s)(s-1)^2] & E_s[e(s)s(s-1)]/\sqrt{2} \\ E_s[e(s)s(s-1)]/\sqrt{2} & E_s[a(s)s^2/2] \end{pmatrix}, \quad (6)$$

$$G_3 = \begin{pmatrix} E_s[ap_5] & E_s[e(s)s^2(s-2)]/\sqrt{12} \\ E_s[e(s)s^2(s-2)]/\sqrt{12} & E_s[ap_4] \end{pmatrix}. \quad (7)$$

For a real symmetric matrix, $\lambda_{\max}$ denotes its largest eigenvalue.

Lemma 2. *If all the following quantities are at most one,*

$$\begin{aligned} \beta E_s[a], \quad \beta E_s[s/q_R], \quad \beta E_s[s/q_I], \quad \beta \lambda_{\max}(G_2), \\ \beta E_s[p_5/q_R], \quad \beta E_s[p_5/q_I], \quad \beta \lambda_{\max}(G_3), \end{aligned} \quad (8)$$

then, in every dimension and for every $k \in L^2(\gamma_d; \mathbb{C})$,

$$\beta \|C_3 k\|_2^2 \leq \mathcal{Q}(k). \quad (9)$$

An infinite right-hand side is allowed.

Proof. Regard complex L^2 as a real Hilbert space with inner product $\text{Re}\langle \cdot, \cdot \rangle$. The inverse pointwise quadratic form is

$$\mathcal{Q}^{-1}(v) = \int \left[a(s)|v|^2 + e(s) \text{Re}(e^{-2i\phi} v^2) \right] d\gamma_d.$$

Use real orthonormal Hermite bases in the transverse coordinates and the exact-degree decomposition

$$\mathcal{H}_3(\mathbb{C}^d) = \bigoplus_{j=0}^3 \mathcal{H}_j(\mathbb{C}) \otimes \mathcal{H}_{3-j}(\mathbb{C}^{d-1}).$$

Different transverse basis elements, including those of different degrees, are orthogonal in both the conjugate and nonconjugate products. Consequently it suffices to analyze each first-coordinate degree. In that coordinate, orthonormal bases are

$$\begin{aligned} j=0: & \quad 1, \\ j=1: & \quad z_1, \overline{z_1}, \\ j=2: & \quad z_1^2/\sqrt{2}, s-1, \overline{z_1}^2/\sqrt{2}, \\ j=3: & \quad z_1^3/\sqrt{6}, z_1(s-2)/\sqrt{2}, \overline{z_1}(s-2)/\sqrt{2}, \overline{z_1}^3/\sqrt{6}. \end{aligned}$$

Their orthogonality and normalizations follow from angular integration and $E_s[s^r] = r!$.

Write each basis function as $R_m(s)e^{im\phi}$. The term $a|v|^2$ is diagonal in this basis. The second term retains exactly the pairs of angular characters satisfying $m+n=2$. In degree zero this leaves only the diagonal entry $E_s[a]$. In degree one the self-pair $m=1$ has real and imaginary eigenvalues $E_s[s/q_R]$ and $E_s[s/q_I]$; the $m=-1$ entry is their average. In degree two the pair $(0,2)$ gives G_2 on real coefficients and the same matrix with negated off-diagonal entries on imaginary coefficients. These matrices have identical eigenvalues; the unpaired $m=-2$ entry is a diagonal entry of G_2 . In degree three the self-pair $m=1$ gives $E_s[p_5/q_R]$ and $E_s[p_5/q_I]$, while the pair $(-1,3)$ gives G_3 and its off-diagonal sign reversal. The unpaired $m=-3$ entry is a diagonal entry of G_3 .

Thus (8) implies $\mathcal{Q}^{-1}(v) \leq \beta^{-1}\|v\|_2^2$ throughout the range of C_3 . Weighted Cauchy–Schwarz gives

$$|\text{Re}\langle k, v \rangle|^2 \leq \mathcal{Q}(k)\mathcal{Q}^{-1}(v) \leq \beta^{-1}\mathcal{Q}(k)\|v\|_2^2.$$

Taking the supremum over unit v in that real Hilbert subspace proves (9). For $d=1$, only the first-coordinate degree three is present; the same sufficient conditions apply. $\square$

4 The midpoint phase relation

For $\mu > 0$, let

$$\Phi_\mu(v) = \begin{cases} v^2/(2\mu), & |v| \leq 2\mu, \\ 2|v| - 2\mu, & |v| \geq 2\mu. \end{cases} \quad (10)$$

Equivalently, $\Phi_\mu(v) = \max_{|r| \leq 1} (2vr - 2\mu r^2)$.

Lemma 3. For $q_R \geq q_I > 0$, $\lambda > 0$, and real v ,

$$\begin{aligned} & \sup_{\substack{|h|^2 + |k|^2 = 1 \\ \operatorname{Re}(h\bar{k}) = 0}} [2v \operatorname{Re} h - 2\lambda|h|^2 + q_R(\operatorname{Re} k)^2 + q_I(\operatorname{Im} k)^2] \\ &= \max\{q_R, q_I + \Phi_{\lambda+q_I/2}(v)\}. \end{aligned} \quad (11)$$

Proof. By symmetry take $v \geq 0$. Write $h = \rho e^{i\theta}$, $k = \pm i\sqrt{1-\rho^2}e^{i\theta}$. At $\rho = 0$, the angle of k is unrestricted. With $x = \cos \theta$, the expression to maximize is

$$2v\rho x - 2\lambda\rho^2 + (1-\rho^2)[q_R + (q_I - q_R)x^2], \quad 0 \leq \rho \leq 1,$$

and it suffices to take $0 \leq x \leq 1$. If $q_R = q_I$, its maximum occurs at $x = 1$, yielding (11). For $d_0 = q_R - q_I > 0$, the maximizing angle at $\rho < 1$ satisfies

$$x = \min \left\{ \frac{v\rho}{d_0(1-\rho^2)}, 1 \right\}.$$

On the first branch, putting $w = \rho^2$, the value is

$$q_R - (2\lambda + q_R)w + \frac{v^2 w}{d_0(1-w)}.$$

This is convex in w , and its branch condition $v\sqrt{w} \leq d_0(1-w)$ defines an interval starting at zero. Its maximum is therefore attained at zero or at the boundary with $x = 1$. On the second branch the value is $q_I + 2v\rho - (2\lambda + q_I)\rho^2$. This proves the asserted upper bound. Its first term is attained at $h = 0, k = 1$; its second term is attained by real h and purely imaginary k . The endpoint cases follow directly or by continuity. $\square$

For an auxiliary parameter $0 < b < 1/2$ satisfying $2b^2 < \beta$, put

$$c = 1 - 2b^2/\beta > 0, \quad A_0 = 1 - 2b > 0, \quad V(s) = \sqrt{s}(A_0 + bs). \quad (12)$$

Proposition 4. Suppose the positive weights satisfy (8) and $E_s[q_R] < \infty$. For $u \geq 0$, define

$$\mathcal{B}(u) = \lambda - cu + E_s \max\{q_R(s), q_I(s) + \Phi_{\lambda+q_I(s)/2}(\sqrt{u}V(s))\}. \quad (13)$$

Then in every dimension,

$$\|T\|_{\infty \rightarrow 1} \leq \sup_{0 \leq u \leq 1} \mathcal{B}(u). \quad (14)$$

Proof. Let f, g be measurable phase functions, and set $h = (f + g)/2$, $k = (f - g)/2$. Then $|h|^2 + |k|^2 = 1$ and $\operatorname{Re}(h\bar{k}) = 0$ pointwise. Self-adjointness gives the exact identity

$$\begin{aligned} \operatorname{Re}\langle f, Tg \rangle &= \|Ph\|_2^2 - \|Pk\|_2^2 - \lambda(\|h\|_2^2 - \|k\|_2^2) \\ &\quad - \beta(\|C_3h\|_2^2 - \|C_3k\|_2^2). \end{aligned}$$

Let $t = \|Ph\|_2$, and use a unitary coordinate change to write $Ph = tz_1$; when $t = 0$, choose any first coordinate. Applying Lemma 2 and discarding $-\|Pk\|_2^2$ yields

$$\operatorname{Re}\langle f, Tg \rangle \leq \lambda + t^2 - 2\lambda\|h\|_2^2 - \beta\|C_3h\|_2^2 + \mathcal{Q}(k).$$

The function $\psi = z_1(s - 2)/\sqrt{2}$ is a unit vector in the range of C_3 , so completing a square gives

$$-\beta \|C_3 h\|_2^2 \leq \frac{2b^2 t^2}{\beta} + 2\sqrt{2}bt \operatorname{Re}\langle h, \psi \rangle.$$

Also $t^2 = 2t \operatorname{Re}\langle h, z_1 \rangle - t^2$. These formulas reduce the bound to $\lambda - ct^2$ plus the integral of

$$2tV(s) \operatorname{Re}(e^{-i\phi}h) - 2\lambda|h|^2 + q_R(\operatorname{Re}(e^{-i\phi}k))^2 + q_I(\operatorname{Im}(e^{-i\phi}k))^2.$$

Lemma 3 bounds this integral by the one in (13). Since $t \leq \|h\|_2 \leq 1$, the asserted real-part bound follows. No pointwise alignment of h with z_1 is assumed.

Multiplying one original function by a constant phase gives the absolute-value bound. Finally, every function in the complex unit disk is the average of two phase functions: pointwise use

$$re^{i\theta} = \frac{1}{2}e^{i\theta}(r + i\sqrt{1-r^2}) + \frac{1}{2}e^{i\theta}(r - i\sqrt{1-r^2}), \quad 0 \leq r \leq 1.$$

This decomposition is measurable, with an arbitrary phase at zero. It extends the estimate to both L^∞ unit balls, proving (14). $\square$

Corollary 5. Fix λ, β , and let $\mathcal{I}$ be a finite set of triples $(b_j, q_{R,j}, q_{I,j})$, each satisfying the hypotheses of Proposition 4. Let $\mathcal{B}_j$ be its function (13). For the single operator (2),

$$\|T\|_{\infty \rightarrow 1} \leq \sup_{0 \leq u \leq 1} \min_{j \in \mathcal{I}} \mathcal{B}_j(u). \quad (15)$$

Proof. For a fixed pair of phase functions, first choose one global phase rotation making its pairing real and nonnegative. Form its midpoint functions, perform the coordinate alignment in the preceding proof, and set $u = \|Ph\|_2^2$. These choices are made once. The proof of Proposition 4 then bounds this same real pairing by $\mathcal{B}_j(u)$ for every $j \in \mathcal{I}$. Thus it is bounded by their minimum at this same u . Taking the supremum over phase functions and using the measurable phase decomposition from that proposition proves (15). $\square$

The auxiliary triple may therefore be selected after u is known. Only a choice among valid inequalities is made; the parameters λ, β and the operator T are identical in every inequality. Below one triple is fixed on each entire parameter interval, so convexity can be applied to that interval using the same triple at both endpoints.

5 The rational family and its Gram conditions

The index set of the family is

$$\mathcal{I} = \{12, 29, 32, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46\}.$$

Write $b_j = B_j/10^8$, with integer numerators specified by

j	B_j	j	B_j
12	676510	39	8771813
29	4210412	40	9162379
32	5558035	41	9540392
34	6558419	42	9965338
35	7059058	43	10352508
36	7516257	44	10727882
37	7993148	45	11098212
38	8408096	46	11429863

These parameters satisfy $0 < b_j < 1/2$ and $2b_j^2 < \beta$. The accompanying data file `family.json` gives positive integers $R_{j,i}, I_{j,i}, j \in \mathcal{I}, 0 \leq i \leq 12000$. Define

$$(q_{R,j}(s), q_{I,j}(s)) = \begin{cases} (R_{j,i}, I_{j,i})/10^8, & i/500 \leq s < (i+1)/500, \quad i < 12000, \\ (R_{j,12000}, I_{j,12000})/10^8, & s \geq 24. \end{cases} \quad (16)$$

Every entry satisfies $12576853 \leq I_{j,i} \leq R_{j,i} \leq 881396009$. Thus the weights and their inverses are bounded. The SHA-256 identifier of the data file is

d0d7b6795988bf9846f3dffd55007ddaa906db45edeef1b830cba22570c1f9fe.

This finite integer table specifies all weights. The certificate uses only these integers and the rational parameters; no optimization or interpolation procedure is part of their definition or verification.

Lemma 6. *Each of the 16 weight pairs (16), with the common parameter β in (1), satisfies all seven conditions (8) strictly.*

Proof. For $r \leq 3$, use the exact identity

$$\int_a^\infty s^r e^{-s} ds = e^{-a} \sum_{k=0}^r \frac{r!}{k!} a^k. \quad (17)$$

Subtracting at successive endpoints gives the cell moments. The last tail moment is used in full. All scalar and matrix-diagonal entries in (8) are rational combinations of these moments. The squared off-diagonal entries in G_2, G_3 have rational factors $1/2, 1/12$.

The program `verify_gram.py` evaluates these moments with the rational interval arithmetic in `arithmetic.py`. It encloses $e^{-1/500}$ between the consecutive Taylor sums through degrees 61 and 60, respectively, and obtains the endpoint exponentials by interval multiplication. Every rational conversion, multiplication, and division rounds outward onto $10^{-90}\mathbb{Z}$, with integer endpoints. All sums and differences are also enclosed outwardly. There is no numerical quadrature in this calculation.

Across all family members, each of the five scalar quantities in (8) and each diagonal entry of $\beta G_2, \beta G_3$ is less than

$$0.999989989464410785 < 1.$$

For every family member the determinant checks give

$$\begin{aligned} \det(I - \beta G_2) &> 0.053824287051790764, \\ \det(I - \beta G_3) &> 0.105885670985068696. \end{aligned}$$

The diagonal entries of each $I - \beta G_j$ are positive, and its determinant is positive. Thus these matrices are positive definite, which proves the two eigenvalue conditions in (8). Every displayed decimal endpoint denotes an exact rational number. $\square$

6 A certified bound on the entire scalar parameter range

6.1 Convex tangent bounds

Fix one family member and suppress its index for the moment. For fixed s , put $a_s = 2\lambda + q_I(s)$. The function $u \mapsto \Phi_{a_s/2}(\sqrt{u} V(s))$ is concave on $[0, \infty)$: it is linear up to its transition and then a

positive multiple of $\sqrt{u}$ plus a constant, with matching derivatives at the transition. For any $\tau > 0$, its tangent at $u = \tau^2$ is

$$L_{\tau,u}(V) = \begin{cases} uV^2/a_s, & \tau V \leq a_s, \\ (\tau + u/\tau)V - a_s, & \tau V \geq a_s. \end{cases} \quad (18)$$

This is a global upper bound for the concave function. For fixed $u \geq 0$, the two branches agree at their transition and together form a nondecreasing function of $V \geq 0$.

On $[24, \infty)$ we instead use

$$\max\{q_R, q_I + \Phi_{a_s/2}(\sqrt{u}V)\} \leq q_R + q_I + uV^2/a_s. \quad (19)$$

Indeed $\Phi_{a_s/2}(v) \leq v^2/a_s$ on both branches; on the linear branch the difference is $(|v| - a_s)^2/a_s$. Define

$$\begin{aligned} J_\tau(u) &= \lambda - cu + \int_0^{24} e^{-s} \max\{q_R, q_I + L_{\tau,u}(V(s))\} ds \\ &\quad + \int_{24}^\infty e^{-s} (q_R + q_I + uV(s)^2/a_s) ds. \end{aligned} \quad (20)$$

Then $\mathcal{B}(u) \leq J_\tau(u)$. Moreover, J_τ is convex in u : each integrand on $[0, 24]$ is the maximum of two affine functions of u , and the remaining terms are affine. Consequently,

$$\sup_{u \in [l, r]} \mathcal{B}(u) \leq \max\{J_\tau(l), J_\tau(r)\}. \quad (21)$$

Restore the family index, writing $J_{j,\tau}$ for (20) with the data of member j . Use exactly the 64 intervals $[k/64, (k+1)/64]$, with

$$\tau_k = 10^{-6} \left\lfloor 10^6 \sqrt{\frac{2k+1}{128}} \right\rfloor, \quad 0 \leq k < 64. \quad (22)$$

Assign the following single member to the entire k -th interval:

$$\sigma(k) = \begin{cases} 12, & 0 \leq k \leq 20, \\ 29, & 21 \leq k \leq 29, \\ 32, & 30 \leq k \leq 32, \\ 34, & 33 \leq k \leq 34, \\ k, & 35 \leq k \leq 45, \\ 46, & 46 \leq k \leq 63. \end{cases} \quad (23)$$

All τ_k are positive rationals. A tangent at this rational point suffices for (21); it need not be taken at the exact midpoint. By (21) and Corollary 5,

$$\|T\|_{\infty \rightarrow 1} \leq \max_{0 \leq k < 64} \max \{J_{\sigma(k), \tau_k}(k/64), J_{\sigma(k), \tau_k}((k+1)/64)\}. \quad (24)$$

Indeed, on any one interval the minimum in (15) is bounded above by its selected $\mathcal{B}_{\sigma(k)}$, and convexity is applied with that fixed member and that fixed tangent. The 128 endpoint bounds in (24) consequently control every $u \in [0, 1]$. The assignment (23) and the tangent parameters are part of the finite data; their validity follows from the exact inequalities below, without any assumption about how they were chosen.

6.2 Exact upper integration

Write $F(s) = V(s)^2 = s(A_0 + bs)^2$. The functions V, F increase on $[0, \infty)$, since $A_0, b > 0$. On a weight cell $[l, r]$, the values q_R, q_I, a_s are positive rational constants.

If $\tau^2 F(r) \leq a_s^2$, the whole cell uses the quadratic branch of (18). If $uF(r) \leq a_s(q_R - q_I)$, the maximum in (20) equals q_R throughout the cell. If $uF(l) \geq a_s(q_R - q_I)$, it equals $q_I + uF/a_s$ throughout the cell. Both cases are integrated exactly by (17).

If $\tau^2 F(l) \geq a_s^2$, the whole cell uses the linear branch. Put $L = \tau + u/\tau > 0$. The maximum is constantly q_R when $L^2 F(r) \leq (q_R + 2\lambda)^2$, and equals $LV(s) - 2\lambda$ throughout the cell when $L^2 F(l) \geq (q_R + 2\lambda)^2$. All squared comparisons have nonnegative sides before squaring.

For the latter case choose the positive rational number

$$r_0 = 10^{-8} \left\lfloor 10^8 \sqrt{(l+r)/2} \right\rfloor.$$

The global inequality $\sqrt{s} \leq (s + r_0^2)/(2r_0)$, multiplied by $A_0 + bs > 0$, gives the polynomial upper bound

$$V(s) \leq \frac{A_0 r_0}{2} + \left(\frac{A_0}{2r_0} + \frac{br_0}{2} \right) s + \frac{b}{2r_0} s^2. \quad (25)$$

Its coefficients are rational, and multiplication by $L > 0$ followed by (17) yields the required integral upper bound.

For a cell crossing a tangent or maximum transition, subdivide its length $h = 1/500$ into 16 equal subintervals. Throughout the original cell the weights remain fixed. On each subinterval, monotonicity in V bounds the integrand by the right-endpoint value of $\max(q_R, q_I + L_{\tau,u}(V))$. If $m_0 = \int_l^r e^{-s} ds$, the exact mass of its subinterval indexed by $v \in \{0, \dots, 15\}$ is

$$m_0 \frac{e^{-vh/16}(1 - e^{-h/16})}{1 - e^{-h}}. \quad (26)$$

Every factor in this expression is outwardly enclosed using $e^{-1/500}$ and $e^{-1/8000}$. Both exponentials are enclosed between their Taylor sums through degrees 61 and 60. This expression for the subinterval mass follows by integrating the exponential over $[l + vh/16, l + (v+1)h/16]$; it does not assume that the exponential density is constant.

At a rational subinterval endpoint s_* , the branch is decided by the exact comparison $\tau^2 F(s_*) \leq a_s^2$. A quadratic endpoint value is rational. A linear endpoint value uses a rational upper bound on $V(s_*) = \sqrt{F(s_*)}$, obtained by rounding an integer square root upward to a multiple of 10^{-24} . Monotonicity gives the upper estimate even when a subinterval itself contains one or more transitions. The tail in (20) is integrated in full by (17), since its weights are constant and V^2 is cubic.

All square-root decisions are exact integer squaring comparisons. The choices τ_k, r_0 need only be positive rational numbers; their downward rounding does not affect the tangent or polynomial inequalities. The endpoint upper bound for $V(r)$ is rounded upward as required. The interval arithmetic for moments is the same outward arithmetic used in Lemma 6.

Lemma 7. *For the family (16) and the fixed assignment (23), the 128 endpoint upper integrals in (24) are all less than $D = 3447/5000$. In particular,*

$$\|T\|_{\infty \rightarrow 1} \leq \frac{3447}{5000} \quad (27)$$

in every dimension.

Proof. The program `verify_scalar.py`, using `scalar_kernel.py` and the integer interval arithmetic in `arithmetic.py`, performs the classifications and integrations above. It verifies the data identifier, the complete 64-interval partition, the validity of each selected member, and the occurrence of both endpoints of each interval exactly once with its prescribed member and tangent. Every endpoint integral in (24) is bounded above by an explicitly integrated majorant. Among the 1,536,000 original radial cell evaluations, 7,304 use the 16 subdivisions in (26); all other cells use their polynomial or constant upper integrands directly.

The largest upper enclosure is less than

$$0.689291686947801369 < \frac{3447}{5000}.$$

It occurs for member 41, interval $k = 41$, right endpoint $u = 21/32$, and $\tau = 100657/125000$. The margin below D is greater than 0.000108313052198631. These numbers bound the computed upper majorants; no lower bound on $\mathcal{B}_j$ or the operator norm is inferred from them. Equation (24) now proves the bound for every parameter in $[0, 1]$, and Lemma 6 verifies every required projection inequality. The programs use Python integers and exact fractions only. No numerical quadrature, numerical eigenvalue calculation, or numerical maximization is used in the verification. $\square$

7 A feasible vector assignment

For Z distributed according to γ_d , let $R = \|Z\|$, $\mu_d = \mathbb{E}R$, and define $F(z) = z/\|z\|$ off the null set $z = 0$. The vector function F has unit norm almost everywhere.

Lemma 8. *For the operator (2), the assignment F has value*

$$N_d := \sum_{j=1}^d \langle F_j, T F_j \rangle = \frac{\mu_d^2}{d} \left(1 - \frac{\beta}{4(d+1)} \right) - \lambda. \quad (28)$$

For the parameters (1),

$$N_d \geq \frac{2d}{2d+1} \left(1 - \frac{\beta}{4(d+1)} \right) - \lambda, \quad N_d \longrightarrow 1 - \lambda. \quad (29)$$

Proof. Write $Z = RU$, where U is uniform on the complex unit sphere and independent of R . Gaussian polar integration gives

$$\mathbb{E}R^2 = d, \quad \mathbb{E}R^3 = (d + \tfrac{1}{2})\mu_d, \quad \mathbb{E}|U_j|^2 = \frac{1}{d},$$

and

$$\mathbb{E}|U_j|^4 = \frac{2}{d(d+1)}, \quad \mathbb{E}|U_j|^2|U_k|^2 = \frac{1}{d(d+1)} \quad (j \neq k).$$

For example, the sphere moments follow by writing the independent exponential random variables $|Z_j|^2$ as their sum times the normalized coordinates; the normalized coordinates have uniform simplex density. Coordinatewise phase rotations show that $\langle F_j, z_k \rangle = \delta_{jk}\mu_d/d$. Hence $\sum_j \|P F_j\|_2^2 = \mu_d^2/d$.

Under coordinatewise phase rotations, F_j has character e_j . A degree-three complex Hermite polynomial with this character has bidegrees $(2, 1)$. The possible orthonormal basis elements are exactly

$$B_j = z_j(|z_j|^2 - 2)/\sqrt{2}, \quad B_{jk} = z_j(|z_k|^2 - 1) \quad (k \neq j).$$

Indeed, its holomorphic and antiholomorphic multiindices satisfy $\alpha - \delta = e_j$, $|\alpha| + |\delta| = 3$, so $\delta = e_k$, $\alpha = e_j + e_k$. Gaussian moments give

$$\begin{aligned}\langle F_j, B_j \rangle &= \frac{1}{\sqrt{2}} \left(\frac{2(d+1/2)\mu_d}{d(d+1)} - \frac{2\mu_d}{d} \right) = -\frac{\mu_d}{\sqrt{2}d(d+1)}, \\ \langle F_j, B_{jk} \rangle &= \frac{(d+1/2)\mu_d}{d(d+1)} - \frac{\mu_d}{d} = -\frac{\mu_d}{2d(d+1)}.\end{aligned}$$

Squaring and summing over all coordinates gives

$$\sum_j \|C_3 F_j\|_2^2 = d \left(\frac{\mu_d^2}{2d^2(d+1)^2} + (d-1) \frac{\mu_d^2}{4d^2(d+1)^2} \right) = \frac{\mu_d^2}{4d(d+1)}.$$

Since $\sum_j \|F_j\|_2^2 = 1$, this proves (28). Finally, Cauchy–Schwarz gives

$$d^2 = (\mathbb{E}R^2)^2 \leq (\mathbb{E}R)(\mathbb{E}R^3) = (d + \tfrac{1}{2})\mu_d^2, \quad \mu_d^2 \leq d.$$

The positive factor $1 - \beta/[4(d+1)]$ yields (29) and the stated limit. $\square$

8 An explicit finite matrix

We now fix

$$d = 10^9, \quad L = 10^9, \quad \Delta = 2^{-80}. \quad (30)$$

In each of the $2d$ standard real coordinates, partition $[-L, L]$ into consecutive intervals of length Δ , and add the two exterior intervals. Assign endpoints consistently; their Gaussian measures are zero. Let $\mathcal{P}$ be the finite product partition, whose size is $(2L2^{80} + 2)^{2d}$.

For two standard real vectors $w, w' \in \mathbb{R}^{2d}$, write $a = w \cdot w'$, $r = \|w\|^2$, $s = \|w'\|^2$. The real degree-three projection kernel is

$$H_3(w, w') = \frac{a^3 - 3a(r+s) + 3(2d+2)a}{6}. \quad (31)$$

To verify the formula, sum the products of degree-three normalized Hermite polynomials, separating the multiindices of types (3), (2, 1), and (1, 1, 1). Their contributions are respectively

$$\sum_i \frac{(w_i^3 - 3w_i)(w_i'^3 - 3w_i')}{6}, \quad \sum_{i \neq j} \frac{(w_i^2 - 1)(w_i'^2 - 1)w_j w_j'}{2}, \quad \sum_{i < j < k} w_i w_j w_k w_i' w_j' w_k'.$$

Expanding these expressions gives (31).

For cells $A, B \in \mathcal{P}$, define the complex matrix

$$M_{AB} = \int_{A \times B} \left(\sum_{j=1}^d \bar{z}_j v_j - \beta H_3(w, w') \right) d\gamma_d(z) d\gamma_d(v) - \lambda \gamma_d(A) \mathbf{1}_{A=B}. \quad (32)$$

Here w, w' are the standardized real coordinates corresponding to z, v . All entries are explicit finite sums of products of one-dimensional Gaussian interval moments, with the displayed diagonal mass term. In particular, (32) specifies a finite matrix without a limiting definition.

Lemma 9. *The matrix (32) satisfies*

$$S_{\mathbb{C}}(M) \leq D, \quad V_{\mathbb{C}}(M) \geq Q_d := \frac{2d}{2d+1} \left(1 - \frac{\beta}{4(d+1)}\right) - \lambda - 2 \cdot 10^{-12}. \quad (33)$$

Proof. The coefficient kernel in (32) has the conjugation appropriate to a first-linear inner product. Directly,

$$\langle f, Pg \rangle = \iint \left(\sum_j \bar{z}_j v_j \right) f(z) \overline{g(v)} d\gamma_d(z) d\gamma_d(v).$$

The kernel H_3 is real symmetric, and the identity term on step functions is $\sum_A \gamma_d(A) f_A \overline{g_A}$. Thus every pair of cellwise phase assignments satisfies

$$\sum_{A,B} M_{AB} \alpha_A \bar{\beta}_B = \langle f, Tg \rangle.$$

Corollary 5 and Lemma 7 prove the scalar bound in (33) for all phases.

For each bounded cell, let z_A be its complex midpoint and put $F_{\mathcal{P}}(A) = z_A / \|z_A\|$; for every exterior cell put $F_{\mathcal{P}}(A) = 0$. A midpoint is never zero, since every real coordinate midpoint is an odd multiple of 2^{-81} . On a bounded cell, $\|z - z_A\| \leq \sqrt{d} \Delta / 2$. For $\|z\| \geq 1$, the normalization inequality gives

$$\|F(z) - F_{\mathcal{P}}(z)\| \leq 2\|z - z_A\| / \|z\| \leq \sqrt{d} \Delta.$$

On the remaining bounded cells the difference is at most two, while on exterior cells its norm is one almost everywhere. Markov's inequality gives

$$\Pr(\|Z\| < 1) \leq e \mathbb{E} e^{-\|Z\|^2} = e 2^{-d}.$$

The standard real Gaussian fourth moment is three, so a union bound gives exterior probability at most $6d/L^4$. Therefore

$$\|F - F_{\mathcal{P}}\|_2^2 \leq d 2^{-160} + 4e 2^{-d} + 6d/L^4 < 10^{-24}. \quad (34)$$

For the last comparison, the first term is below 10^{-30} because $d 10^{30} < 2^{160}$. The second is below 10^{-30} using $e < 3$, $d \geq 200$, and $12 \cdot 10^{30} < 2^{200}$. The third is $6 \cdot 10^{-27}$.

The ranges of P and C_3 are orthogonal. The eigenvalues of T are $1 - \lambda$, $-\lambda - \beta$, and $-\lambda$, on their respective orthogonal subspaces. Since $\lambda + \beta = 553209/10^6 < 1$,

$$\|T\|_{2 \rightarrow 2} = \max(1 - \lambda, \lambda + \beta) < 1.$$

The same norm holds for componentwise action on vector-valued L^2 . Since $\|F\|_2 = 1$, $\|F_{\mathcal{P}}\|_2 \leq 1$, (34) implies

$$\left| \sum_j \langle F_j, T F_j \rangle - \sum_j \langle (F_{\mathcal{P}})_j, T (F_{\mathcal{P}})_j \rangle \right| < 2 \cdot 10^{-12}.$$

Finally assign to the row and column cells the unit vectors

$$u_A = \left(F_{\mathcal{P}}(A), \sqrt{1 - \|F_{\mathcal{P}}(A)\|^2}, 0 \right), \quad v_B = \left(F_{\mathcal{P}}(B), 0, \sqrt{1 - \|F_{\mathcal{P}}(B)\|^2} \right).$$

Their cross inner products are those of $F_{\mathcal{P}}$. The real part of their matrix value is therefore at least Q_d , by Lemma 8. This proves the vector bound. $\square$

Proof of Theorem 1. The scalar constant is $D = 3447/5000$. For the explicit finite parameters (30), the rational numerator lower bound in (33) is

$$Q_d = \frac{315558000264346749990112333333}{333333333833333333500000000000}.$$

An exact rational comparison gives

$$Q_d - \frac{1373}{1000}D = \frac{42599791073649832354633333}{333333333833333333500000000000} > 0.$$

In particular $Q_d > 0$, so $M \neq 0$. Lemma 9 yields $V_{\mathbb{C}}(M)/S_{\mathbb{C}}(M) \geq Q_d/D > 1373/1000 = 1.373$. The corresponding limiting comparison is

$$\frac{1 - \lambda}{D} = \frac{52593}{38300} > 1.373.$$

□

References

- [1] U. Haagerup, *A new upper bound for the complex Grothendieck constant*, Israel Journal of Mathematics **60** (1987), no. 2, 199–224. [doi:10.1007/BF02790792](https://doi.org/10.1007/BF02790792).
- [2] S. Friedland, L.-H. Lim, and J. Zhang, *An elementary and unified proof of Grothendieck’s inequality*, L’Enseignement Mathématique **64** (2018), no. 3/4, 327–351. [doi:10.4171/LEM/64-3/4-6](https://doi.org/10.4171/LEM/64-3/4-6).
- [3] S. Heilman, *A lower bound for Grothendieck’s constant*, [arXiv:2603.22616v1](https://arxiv.org/abs/2603.22616) (2026).